Python 3.15 Compatibility Findings

Author: Dr. Ceasar Jackson Jr.
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Executive Summary

Testing of Python 3.15.0b2 on macOS ARM64 shows that most of the modern data engineering stack is operational. NumPy, Pandas, Polars, DuckDB, SQLAlchemy, Matplotlib, Plotly, Airflow, and MLflow passed validation.

Major blockers remain for SciPy, Prefect, and Dask due to upstream compatibility issues rather than local environment configuration.

Key Findings

SciPy: INCOMPATIBLE

No cp315 wheel is available. Source compilation proceeds successfully through OpenBLAS, CMake, Meson, Ninja, and GFortran validation but fails during Pythran code generation.

```
Observed failure: TypeError: ast.Compare.__init__ missing 1 required positional
argument: 'left'
```

Root cause appears to be an upstream incompatibility involving Python 3.15 AST changes, Pythran, and GAST.

Prefect: INCOMPATIBLE

Current releases depend on typing.no_type_check_decorator, which was removed in Python 3.15.

Dask: INCOMPATIBLE

Validation blocked by unavailable PyArrow support for Python 3.15.

Production Readiness Assessment

Suitable for:

Research, CI validation, migration planning, compatibility testing, and exploratory evaluation.

Not Yet Suitable for:

Production scientific computing workloads requiring SciPy, Dask/PyArrow, or Prefect.

Recommendation

Use Python 3.14.x in production and continue monitoring Python 3.15 ecosystem support.